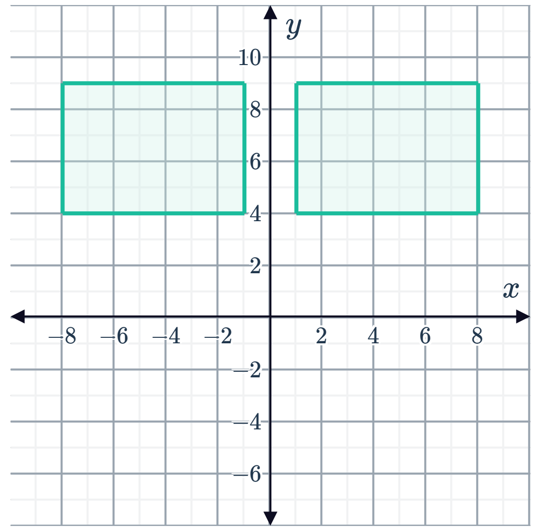
Transformations, congruence, and si[redacted]ity

1 Consider the figures shown.



a Are the two shapes similar, congruent or neither?

b Determine whether the following are the possible types of transformation from one quadrilateral to the other:

i Translation

ii Dilation

iii Reflection

iv Rotation

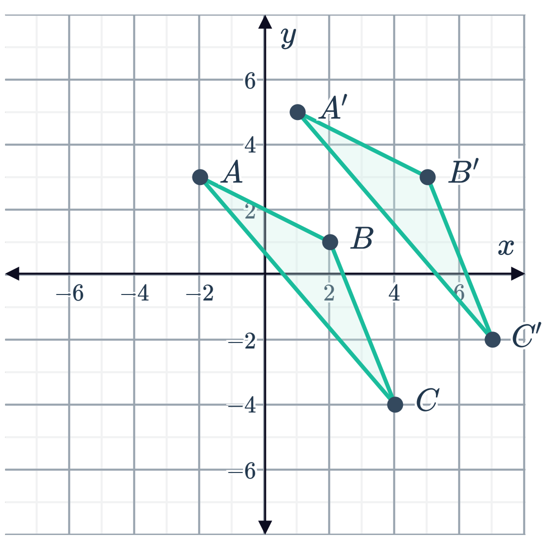
2 Consider the following pairs of triangles:

i Are the two triangles similar, congruent or neither?

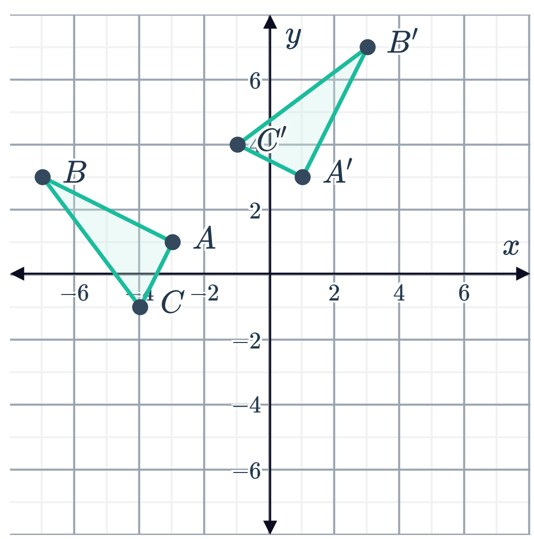
ii Identify the single type of transformation that can take $\triangle ABC$ to $\triangle A'B'C'$.

iii State the transformation or series of transformations from $\triangle ABC$ to $\triangle A'B'C'$.

a



b



c A triangle with vertices at $A(-4,-1)$, $B(1,3)$, and $C(2,-3)$, and another triangle with vertices at $A'(0,-3)$, $B'(5,1)$, and $C'(6,-5)$.

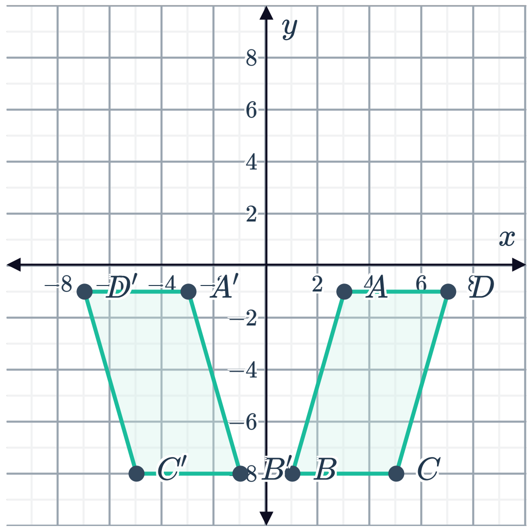
3 Consider the following pairs of quadrilaterals.

i Are the two quadrilaterals similar, congruent or neither?

ii Identify the single type of transformation that can turn quadrilateral $ABCD$ into quadrilateral $A'B'C'D'$.

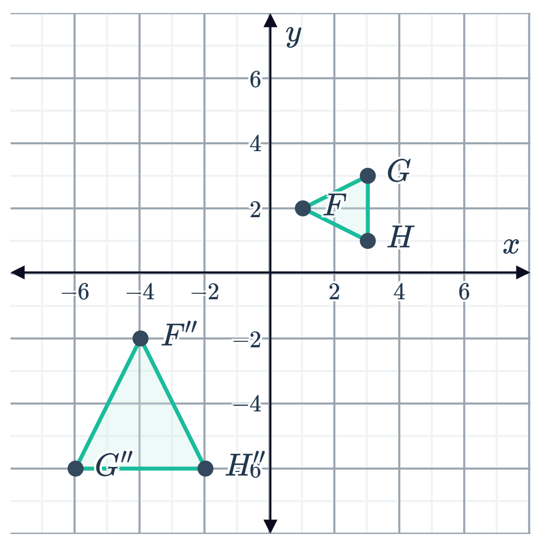
iii State the transformation or series of transformations from from quadrilateral $ABCD$ to quadrilateral $A'B'C'D'$.

a



b A quadrilateral with vertices at $A(2,1)$, $B(2,9)$, $C(8,9)$, and $D(8,1)$, and another quadrilateral with vertices at $A'(1,-2)$, $B'(9,-2)$, $C'(9,-8)$, and $D'(1,-8)$.

4 $\triangle FGH$ and $\triangle F''G''H''$ are shown on the given coordinate plane. State a possible sequence of transformations can be used to show that $\triangle FGH \sim \triangle F''G''H''$.



5 Consider the transformation from (x, y) to $($[redacted]$y - 8)$.

a Determine what type of transformation is this?

b Describe the transformation.

c Is it a congruence transformation or a similarity transformation?

6 Consider the transformation from (x, y) to $(-x, y)$.

a State the transformation.

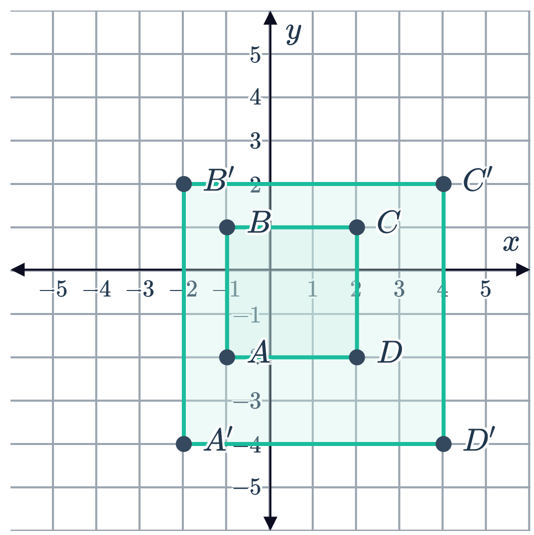
b Is it a congruence transformation or a similarity transformation?

7 Consider a transformation that causes one vertex of a triangle to move from (x, y) to $(0.6x, 0.6y)$.

a State a possible transformation.

b Is it a congruence transformation or a similarity transformation?

8 Consider the coordinate plane shown.



a Are the two rectangles similar, congruent or neither?

b Find the scale factor of the dilation from rectangle $ABCD$ to rectangle $A'B'C'D'$.

9 Consider the triangle with vertices at $A\,(-3, -2)$, $B\,(2, 1)$ and $C\,(3, -3)$, and the triangle with vertices $A'\,(-9, -6)$, $B'\,(6, 3)$ and $C'\,(9, -9)$.

a Are the two triangles similar, congruent or neither?

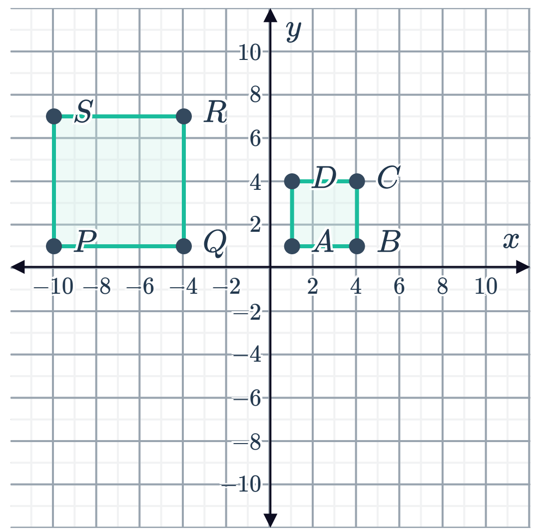
b Find the scale factor of the dilation.

10 Consider the quadrilateral with vertices at [redacted] $-3)$, $B\,(-3,3)$, $C\,(3,3)$ and $D\,(3,-3)$, and the quadrilateral with vertices at $A\,(-9,-9)$, $B\,(-9,9)$, $C\,(9,9)$ and $D\,(9,-9)$.

a Are the two rectangles similar, congruent or neither?

b Find the scale factor of the dilation of rectangle $ABCD$ to rectangle $A'B'C'D'$.

11 Two squares are shown in the following diagram.
Determine whether the following sequences of transformations could be used to show that the two squares are similar.



a Translate square $ABCD$ by 4 units to the right, then reflect it across the y-axis, and then dilate it by a factor of $\frac{1}{2}$ with a centre of dilation at point A''.

b Translate square $ABCD$ by 11 units to the left, then dilate it by a factor of 2 with a centre of dilation of point A'.

c Reflect the square $ABCD$ across the x-axis, translate it 4 units to the right, then dilate it by a factor of 2 with a centre of dilation of point A''.

d Rotate square $ABCD$ by 90° clockwise around the origin, then dilate it by a factor of $\frac{1}{2}$ with a centre of dilation of point A'.

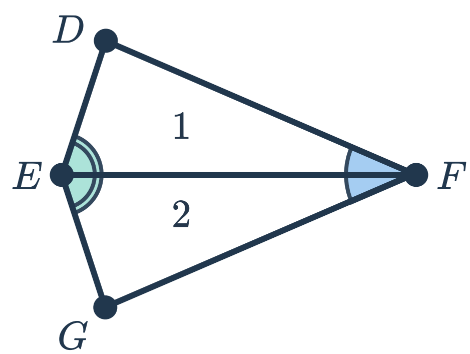
12 An equilateral triangle is rotated about its center.
Determine whether the following angle rotations map the triangle onto itself:

a 270° b 60° c 90° d 120°

e 180° f 240°

Additional questions

13 Consider the pair of congruent triangles shown:



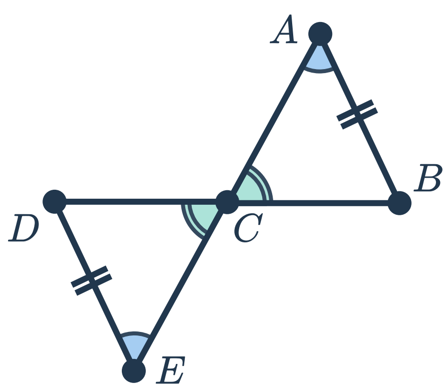
a What transformation maps triangle 1 to triangle 2?

b Write a congruency statement for each pair of sides

c Write a congruency statement for each pair of angles

14 Determine which of the following pairs of shapes are congruent. For each congruent pair, state the transformation or series of transformations that maps one shape to the other.

a



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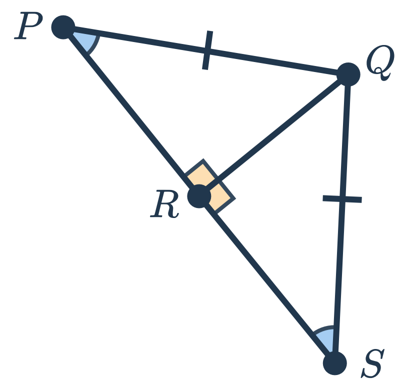

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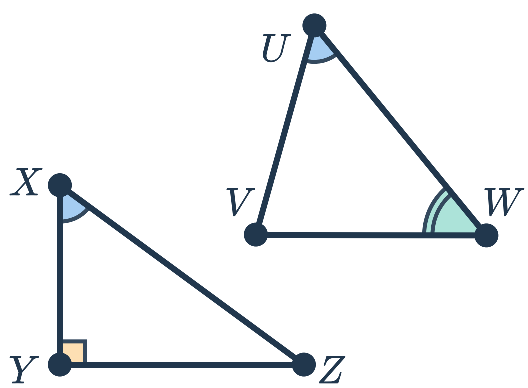

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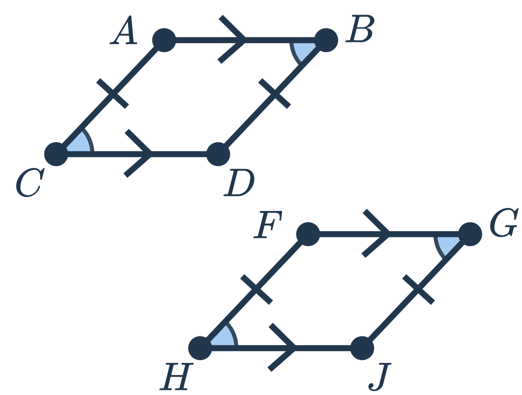

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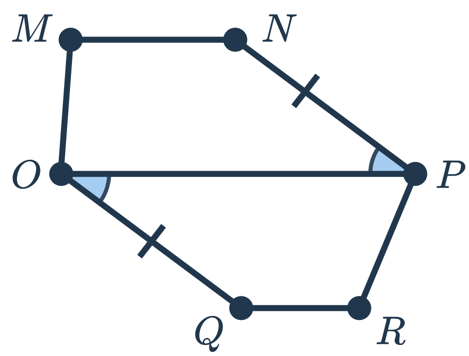
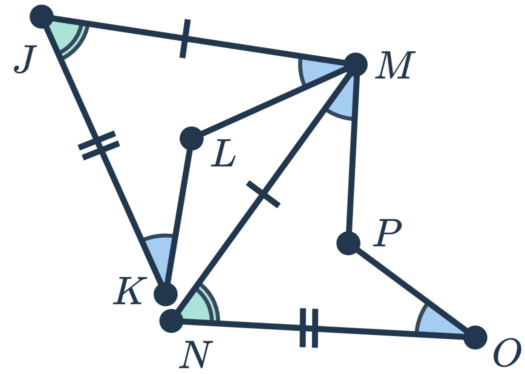




15 Explain why performing a combination of translations, rotations, and reflections will always produce a congruent figure.

16 Give an example of two triangles which have one pair of congruent angles, and two pairs of congruent sides, but are not themselves congruent triangles.

17 Give an example of two quadrilaterals in which all four pairs of sides are congruent, but the quadrilaterals themselves are not congruent.

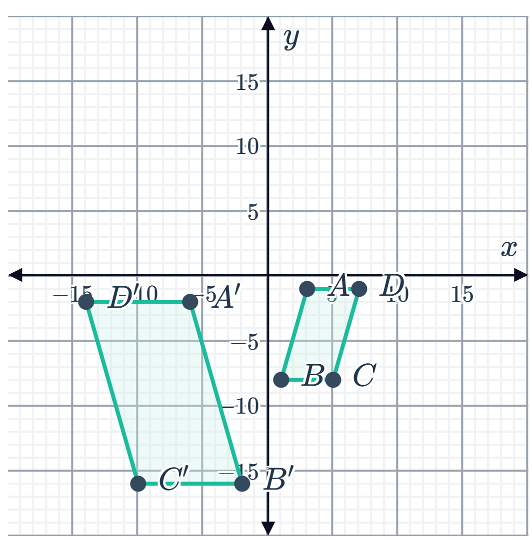
18 For each pair of figures:

a Identify the scale factor of the dilation.

b Describe a transformation or series of transformations that proves these figures are similar.

c Write the transformation rule using function notation.

a



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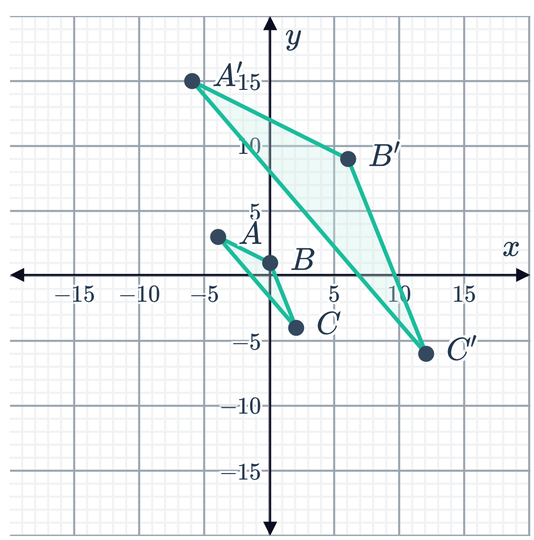

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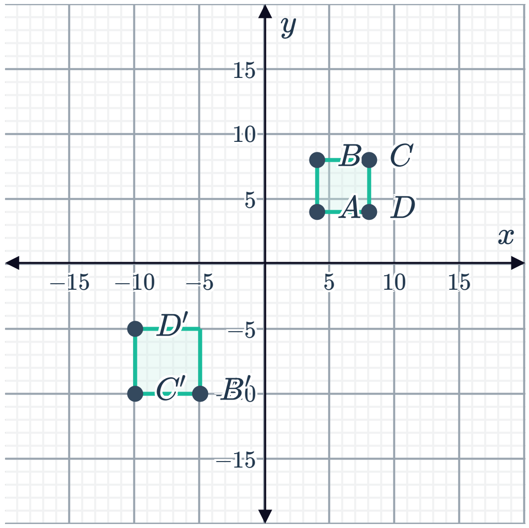
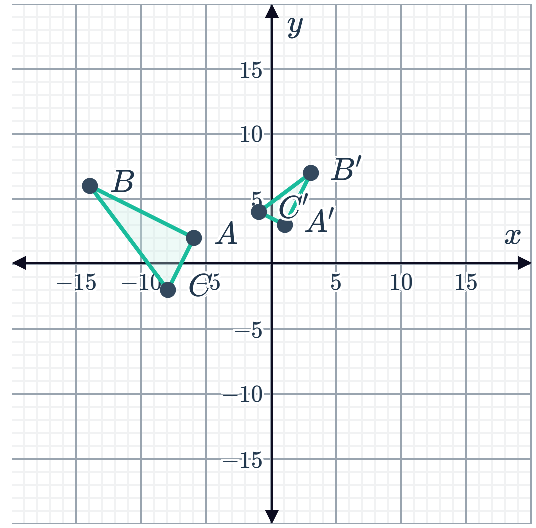
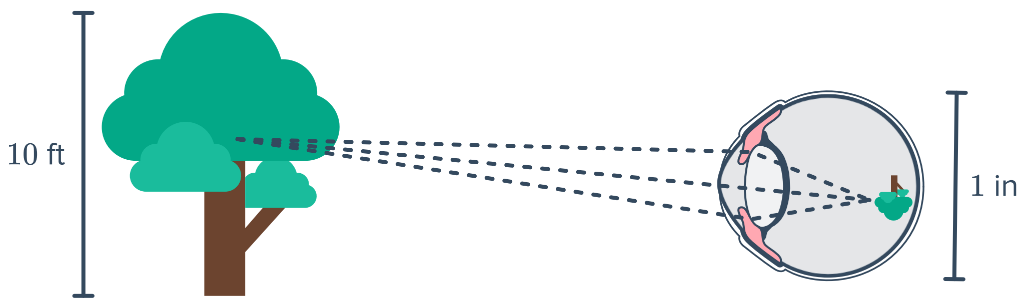




19 In order to see an object, the eye performs a series of transformations as depicted in the image below. These transformations allow the image to project onto the back of the eye before being sent to the brain for processing.
Describe the series of transformations used in order to "see" the tree in the image if the eye is 20 ft from the tree. Let the origin be through the center of the tree.



20 Decide if the following conjectures are true or false. Justify each response.

a Congruent figures are always similar.

b "If two quadrilaterals have congruent angles, then they must be similar."

c "All isosceles triangles are similar."

d "All equilateral triangles are similar."

21 Identify the coordinates of $(12, -12)$ after the sequence of transformations:

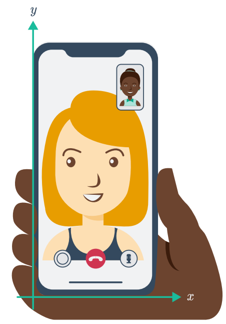
a Dilate by a scale factor of 1.5

b Reflect over the y-axis and then dilate by a scale factor of 2

c Rotate 90° about the origin and then dilate by a scale factor of $\frac{1}{4}$

d Dilate by a scale factor of 3 and then translate using the rule $(x, y) \rightarrow (x - 1, y + 3)$

22 A video phone call includes a full screen image of the person being called and a smaller image of the caller in the corner of the screen (see image for reference). Consider making a video call on a smart phone which is 1782×828 pixels in dimension.



a What considerations need to be made in order to describe the transformations that take the caller's image from the full screen to the smaller screen in the top right corner.

b Using your best estimations, describe a series of transformations in pixels that might be used to make the caller's face a small screen in the top left corner.